

SPIRE

A Contested-Logistics Decision Engine for the Stand-in Force

BLUF. A 3d MLR S-4 forward in the First Island Chain today cannot see Class I, V, and VIII in a single pane — those classes live in three incompatible systems while GCSS-MC focuses on Class IX requisitioning. SPIRE collapses GCSS-MC, DRRS-MC, and forward sensor feeds into one role-shaped view that runs offline on a 2U box in a CONEX, with tamper-evident audit and one-click coalition release. **TRL 4, validated at Modern Day Marine 2026 (1st place, 30 APR 2026). Operational pilot is the next step.**

The Operator's Problem

Today, a forward unit taking a Class VIII demand spike under degraded SATCOM has three systems open and a phone call. The planner has minutes — not hours — to choose between cross-level, expedite, or emergency requisition. The decision is unauditable; the artifact is a spreadsheet.

Capability Gap

“GCSS-MC now primarily focuses on requisitioning Class IX repair parts.” Classes I, V, and VIII are managed by “three separate incompatible systems.” “Conventional sustainment remain[s] the SIF concept’s greatest vulnerability.”

— *Sustainment of the Stand-in Force*, War on the Rocks (Sept 2022). The same fragmentation is acknowledged in the *Marine Corps Concept for Logistics* (Dec 2024).

What SPIRE does

Four role-shaped surfaces over one canonical dataset:

- **Decision Bridge.** Home view — five live signals on one non-scrolling pane: mission clock + FPCON, top alerts, forecasted Class I / V / VIII / IX shortages, mission-capable rate by unit with seven-day sparklines, audit-chain health.
- **PULSE.** Readiness and risk. Monte Carlo demand forecast over 200 paths; auto-drafted ranked actions (cannibalize, expedite, cross-level) with impact-per-day; predicted-failure column drives auto-draft requisitions before assets actually break.
- **BASTION.** Common operating picture on real satellite tiles with MIL-STD-2525D unit symbols; multi-sensor threat fusion (PACS + ThermalHawk + SCADA + weather), QRF cordons, threat rings. A natural-language copilot drives the map.
- **SENTRY.** CUI auto-tagging pipeline. Tier-1 rule engine, opt-in Tier-2 Gemma 4 explainer. Sanitized export and one-click coalition release scoped per partner — FVEY, JPN, AUS, PHL — with manifest hash and release ID stamped to the audit chain.

Under the hood

- **Universal Ingest Service.** 8 formats × 6 channel types (drop, SFTP, IMAP, HTTP, DB CDC, Kafka), 4 pre-built GCSS-MC / DRRS-MC adapters; a local Gemma 4 26B handles schema mapping for everything else. Mappings save per unit.
- **Tamper-evident audit chain.** Every state change SHA-256 hash-chained and Ed25519-signed; CEF export for Splunk / ArcSight / QRadar; inspector verification offline against a published key.
- **Local-first.** Runs on a 2U CONEX box; vector-clock conflict resolution reconciles forward-node writes when SATCOM restores; no cloud egress.

Maturity & Risk

TRL 4. Validated against synthetic GCSS-MC ECP / UTIL / SR-Header data at MDM 2026; hash-chained audit verified offline.

TRL 5 requires a sanitized real GCSS-MC pull-through from a sponsoring MEF G-4 or MLR S-4. ATO is not yet established; architecture is designed-in for IL5 (audit chain, hierarchical RBAC, CAC/PIV scaffolding, no cloud egress) but unaccredited.

Why Now

The 39th Commandant has named logistics the **pacing function**. The *Marine Corps Concept for Logistics* (Dec 2024) calls for resilience through redundancy, dispersion, and innovation. The data-fragmentation gap is named in published doctrine. The window to put SPIRE in operator hands is the FY27 POM cycle.

Path Forward — Ask

1. **90-day operational pilot** with one MEF G-4 or MLR S-4 willing to point a sanitized GCSS-MC export at the ingest path.
2. **An MCWL evaluator** to red-team SENTRY against operator-malpractice patterns.
3. **A POC at MCSC PM-MC2I or DC I&L** to translate operator requirements into a contract vehicle (SBIR Phase II or OTA).

POC: Jesse Morgan · Thornveil · jesse@thornveil.ai

Repo: github.com/jeranaia/spire (public)

Pilot install: `docker-compose up` — under 5 min, clean laptop to live demo

v1.1 · 04 MAY 2026 · Distribution unrestricted by author